CYBERSECURITY · SEC-04

Linux Security Commands

Use this when: you check a Linux box for who has access, what is running and listening, what changed, and how to lock it down.

BreakFixLearn

Security+ · Linux+
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Scan to visit

Read a permission line

r = 4 · w = 2 · x = 1, add them up

-	rwx	r-x	---	
type	owner = 7	group = 5	others = 0	alice ops script.sh

`chmod 750 script.sh` sets exactly that. Type: `-` file, `d` directory, `l` link.
`chmod u+x` adds execute for the owner. `chown alice:ops` sets owner and group.

SAFE DEFAULTS

644	Normal files
755	Directories, scripts
600	Private keys, secrets
700	~/.ssh
4000	SUID: runs as the file owner

Who has access?

`id` · `whoami`	My user, groups
`sudo -l`	What can I sudo?
`w` · `last`	Who is on, logins
`sudo lastb`	Failed logins
`awk -F: '$3==0' /etc/passwd`	UID 0 (root) users

What is running and listening?

`ss -tulpn`	Ports, PIDs
`sudo lsof -i :22`	Port 22 owner
`ps aux --sort=-%cpu | head`	Top CPU
`systemctl --state=running`	Running units

What changed? Find risky files

`find / -xdev -perm -4000 -type f`	SUID files
`find / -xdev -perm -0002 -type f`	World-writable
`find /etc -mtime -1`	Changed in 24 h
`sha256sum file.iso`	Hash to compare

Read the logs

`journalctl -u ssh --since today`	SSH log
`journalctl -p err -b`	Errors, boot
`grep "Failed password" /var/log/auth.log`	SSH guessing

Ubuntu and Debian shown. RHEL: `/var/log/secure`, unit `sshd`.

Lock it down

```
# keys first, from your PC
ssh-keygen -t ed25519
ssh-copy-id admin@192.0.2.10
# firewall (Ubuntu ufw)
sudo ufw default deny incoming
sudo ufw limit 22/tcp   # SSH, rate-limited
sudo ufw enable && sudo ufw status verbose
# /etc/ssh/sshd_config
PermitRootLogin no
PasswordAuthentication no  # keys only
sudo sshd -t && sudo systemctl reload ssh
# patches (RHEL: sudo dnf upgrade)
sudo apt update && sudo apt upgrade
```

Before you press Enter

DANGER `chmod -R 777` and `rm -rf` on the wrong path can't be undone.

CAUTION Allow SSH before `ufw enable`, or a remote session locks you out.

CAUTION Set up SSH keys and keep a second session open before you turn off passwords.

LEGAL Run these only on systems you administer.

REMEMBER

- Use **sudo** per command, not a root shell
- `sshd -t` before every reload
- Patch on a schedule, not when you remember

Sources: man7.org man pages (ss, find, journalctl, sshd, sshd_config) · Debian ufw(8) · CompTIA Security+

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